BIOLOGY | ADAPTLY

CELL TRANSPORT – DIFFUSION & OSMOSIS

KEY VOCABULARY

Diffusion: net movement of particles from high to low concentration.

Osmosis: diffusion of water across a partially permeable membrane.

Active transport: moving substances against the gradient using energy.

Concentration gradient: the difference in concentration between two areas.

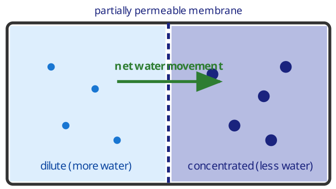
COMMON MISCONCEPTIONS

~~Diffusion needs energy from respiration.~~
✓ Diffusion is passive — it needs no energy.

~~Osmosis is the movement of any substance.~~
✓ Osmosis is the movement of water only.

~~Particles stop moving once balanced.~~
✓ Particles still move, but there is no net change.

OSMOSIS ACROSS A MEMBRANE



Water moves from a **dilute** solution to a **concentrated** one.

1 DEFINE DIFFUSION

Describe what is meant by **diffusion**. **[2 marks]**

2 OSMOSIS

Explain how osmosis is different from diffusion. **[2 marks]**

3 RATE OF DIFFUSION

State **three factors** that increase the rate of diffusion. **[3 marks]**

4 REQUIRED PRACTICAL

Potato cylinders are placed in sugar solution and lose mass. **Explain** why. **[2 marks]**

5 PERCENTAGE CHANGE

A cylinder changes from 5.0 g to 4.0 g. **Calculate** the percentage change in mass. **[2 marks]**

Answer: ______ %

6 ACTIVE TRANSPORT

Complete the sentence using the word bank. **[2 marks]**

Active transport moves substances _____ the concentration gradient and requires _____.

against | energy | with | no energy

TIP: *Osmosis = water only, across a partially permeable membrane.*

✓ **Think: Why do root hair cells use active transport to absorb minerals?**